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भारतीय मानक मसौदा

थियोनिल क्लोराइड — सुरक्षा संहिता

(IS 9744 का पहला पुनरीक्षण)

Draft Indian Standard

Thionyl Chloride — Code of Safety

(First Revision of IS 9744)

ICS 71.060

Chemical Hazards Sectional Committee, CHD 07

Last date for Comments: As per the covering letter

FOREWORD

(Formal clause shall be added later)

Thionyl chloride, also known as sulphurous oxychloride is a pale yellow to red liquid with a sharp odour. Thionyl chloride is used as a catalyst and chlorinating agent in chemical industry; it is also used in organic synthesis, pesticides, engineering and plastic industries.

Thionyl chloride is very toxic, and in the presence of moisture it yields toxic decomposition products, namely, hydrogen chloride and sulphur dioxide. Though no hygienic standard of permissible exposure has been published, a twenty-minute exposure to 17.5 ppm has been found in experiments on cats to be fatal. Thionyl chloride is used in dyestuff industries, pesticides, and as surface active agent. The use of the thionyl chloride in vulcanizing rubber, in wood hardening and in the extraction of gold from its ores is gradually increasing.

The elimination of accidents is vital to public interest. Accidents produce economic and social loss, and impair individual or group productivity. Realization of this loss has led the authorities to devote a good deal of attention to safety education. In any program of safety education, preparation of code of safety is an essential part. Apart from general precautions, some typical precautions are required to be taken and this code of safety lays special emphasis on these points. For proper utilization of the code of safety for thionyl chloride, a knowledge of effects of hazardous substances on biological systems is desirable. This code of safety recommends practices to be followed to ensure safety of personnel engaged in industries where thionyl chloride is produced, stored, handled or used.

This Indian Standard was originally published in 1981. The Committee felt a need to revise this standard with a view to update the standard based on the experience of last four decades and on the currently available data.

In this revision the following changes have been incorporated:

a) The general properties have been incorporated, and modifications have been made to update safety measures for controlling hazards and essential information on symptoms of poisoning, first-aid, medical treatment, storage, handling, labelling and employee safety, fire and explosion hazards, routes of entry, toxicity information, fire prevention and firefighting, and health monitoring; and

b) The title of the standard has been modified as ‘Thionyl Chloride — Code of Safety’.

There is no standard on this subject published by ISO. This standard has been prepared indigenously based on the available data and information.

The various clauses of the standard have been aligned with the format being applied for all Indian Standards on code of safety of chemicals.

The properties of thionyl chloride listed in Clause 4 have been taken from literature and have been included for information only. BIS has published a separate standard IS 11305: 1985 on the requirements and the methods of sampling and test for thionyl chloride intended for industrial purposes.

1 SCOPE

1.1 This standard prescribes general properties of thionyl chloride and the nature of hazards associated with it.

1.2 This standard also covers other essential information such as information on storage, handling, packing, labelling, disposal of waste, cleaning and repair of containers, selection and training of personnel, protective equipment and first-aid.

1.3 This standard does not deal with specification for design of buildings, chemical engineering plants, storage vessels, equipment for operations control and waste disposal.

2 REFERENCES

The Indian standards listed in **Annex A** contain provisions, which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards.

3 TERMINOLOGY

For this standard the definitions given in IS 4155 and IS 4167 shall apply.

4 PROPERTIES

4.1 General

Thionyl chloride is highly toxic in nature, and in the presence of moisture it yields toxic decomposition products such as hydrogen chloride and sulphur dioxide. It also known as sulphurous oxychloride. It is used as a catalyst and chlorinating agent in chemical industry, it is also used in organic synthesis, pesticides, engineering and plastic industries.

4.1.1 *Chemical Name* — Thionyl chloride

4.1.2 *Common Name & Synonyms* — Sulphur oxychloride, sulphurous oxychloride, thionyl dichloride, sulfinyl chloride, sulfurous dichloride.

4.1.3 *Uses*

Thionyl chloride is used as a chlorinating agent and catalyst in the chemical industry. It is widely used in organic synthesis, pesticide manufacture, engineering and plastics industries, and in the dyestuff industry as well as in the

production of surface-active agents. The use of the thionyl chloride in vulcanizing rubber, in wood hardening and in the extraction of gold from its ores is gradually increasing.

4.2 Identification

4.2.1 *Chemical formula* — SOCl_2

4.2.2 *CAS Number* — 7719-09-7

4.2.3 *UN Number* — 1836

4.2.4 *UN Class* — 8 (Corrosive substance)

4.3 Physical Properties

4.3.1 General

Thionyl chloride (SOCl_2) is a volatile, colourless to pale-yellow (occasionally reddish) fuming liquid with a sharp, pungent and suffocating odour. It has a molecular weight of 118.97 g/mol. It has a melting point of about -105°C and boiling point of 78.8°C at 746 mm Hg. Thionyl chloride has a relative density of about 1.640 at $15.5^\circ\text{C}/15.5^\circ\text{C}$ and a vapour density of 4.1 (air = 1). Thionyl chloride reacts violently with water and forms sulphur dioxide and hydrochloric acid.

4.3.2 *Molecular Weight* — 118.97 g/mol.

4.3.3 *Physical State* — liquid.

4.3.4 *Colour* — colourless to pale yellow or yellow to red, clean liquid.

4.3.5 Odour

Thionyl chloride has characteristic pungent odour similar to that of sulphur dioxide; vapour irritates respiratory system, eyes and skin.

4.3.6 *Boiling Point* — 78.8°C at 746 mm Hg.

4.3.6.1 *Technical grade* — 77°C .

4.3.6.2 *100 percent pure material* — 78.8°C at 746 mmHg.

4.3.7 *Melting Point* — -105°C .

4.3.8 *Freezing Point, $^\circ\text{C}$* — -105°C .

4.3.9 *Vapour Density (air=1)* — 4.1.

4.3.10 *Specific Gravity Liquid (water = 1) at 20°C* — 1.638.

4.3.11 *Viscosity at 0°C* — 0.01 mPas.

4.3.11.1 *Viscosity at 30°C* — 0.545 mPas.

4.3.12 *Vapour Pressure at 27°C* — 110 mmHg.

4.3.12.1 *Vapour Pressure at 20°C* — 97 mmHg.

4.3.13 *Refractive Index at 20°C* — 1.517.

4.3.14 Light Sensitivity

Thionyl chloride is light-sensitive and decomposes on exposure to UV radiation or direct sunlight.

4.3.15 *Relative Density $15.5^\circ\text{C}/15.5^\circ\text{C}$* — 1.640.

4.3.15 *Relative Density 30°C* — 1.62.

4.3.16 *Percentage in 'Saturated' Air* — 14.5 at 27°C .

4.3.17 Relative Density of Saturated Air (air = 1) — 1.5 at 27 °C.

4.3.18 Solubility in Water

Thionyl chloride is not water-soluble. It reacts rapidly and exothermically with water, producing sulfur dioxide and hydrochloric acid fumes.

4.3.19 Solubility in other solvents

Thionyl chloride is soluble in benzene, chloroform, toluene, diethyl ether and carbon tetrachloride.

4.4 Chemical Properties

4.4.1 Reactivity

Thionyl chloride reacts violently with water and forms sulphur dioxide and hydrochloric acid. It is also very reactive with ammonia, chloryl perchlorate, dimethyl sulphoxide, linseed oil, quinoline, sodium, 2,4-hexadiyne- 1 -6-diol, o-nitrobenzoyl acetic acid, o-nitro phenyl acetic acid. It may react with metals to release flammable hydrogen gas. Thionyl chloride decomposes into chlorine, sulphur dioxide and sulphur monochloride when heated above 140 °C.

4.4.2 Polymerisation — Not applicable.

4.4.3 Allotrope Formation — Not applicable.

4.4.4 Corrosion Properties

Thionyl chloride is a highly corrosive substance. It reacts violently with water and forms sulphur dioxide and hydrochloric acid. It can corrode various metals, particularly in the presence of moisture, and may lead to the formation of hydrogen gas.

4.5 Fire and Explosion Hazard Properties

4.5.1 Ignition Temperature — Not applicable.

4.5.2 Auto Ignition Temperature — Not applicable.

4.5.3 Flash Point — Not applicable.

4.5.4 Explosive Limits

4.5.4.1 Upper Explosive Limit — Not applicable.

4.5.4.2 Lower Explosive Limit — Not applicable.

4.5.5 Fire Risk

4.5.5.1 Thionyl chloride is non-flammable substance; however, it can cause ignition on contact with combustible materials due to its strong reactivity and heat-releasing decomposition.

4.5.5.2 Thionyl chloride reacts vigorously with water and fumes in moist air. It hydrolyses with water to form sulphur dioxide and hydrochloric acid. When heated above 140 °C, it decomposes to chlorine, sulphur dioxide, and sulphur monochloride.

5 HEALTH HAZARD WITH THIONYL CHLORIDE

5.1 General

Thionyl chloride vapour irritates respiratory system, eyes and skin. The liquid is highly corrosive and causes burns on the skin, eyes and mucous membranes, whenever it comes in contact with these parts. Its ingestion results in severe internal irritation and damage. However, no permanent ill effect is reported.

5.2 Routes of entry

5.2.1 Skin

Thionyl chloride is highly corrosive and causes severe skin burns, redness, pain, blistering, and tissue destruction on contact. It reacts with moisture on the skin, leading to tissue damage, increasing the severity of burns therefore, immediate decontamination is necessary.

5.2.2 Eyes

Exposure of the eyes may cause intense pain, conjunctival damage, corneal burns, and possibly permanent vision loss. therefore, adequate eye protection is essential during handling.

5.2.3 Ingestion

Ingestion of thionyl chloride is extremely hazardous and may causes severe corrosive burns to the mouth, throat, oesophagus, and stomach, which may lead to gastrointestinal perforation and life-threatening effects.

5.2.4 Inhalation

Inhalation of thionyl chloride vapours is highly dangerous. It reacts with moisture in the respiratory tract to form hydrogen chloride (HCl) and sulphur dioxide (SO₂), causing severe irritation, coughing, choking, and inflammation. High concentrations may result in pulmonary oedema, chemical pneumonitis, or respiratory failure.

5.2.5 Long term effects

Repeated or chronic exposure to thionyl chloride fume can lead to respiratory irritation, bronchitis, and reduced lung function. Prolonged skin exposure causes dermatitis and scarring.

5.3 Toxicity Information

Thionyl chloride is a fuming, corrosive, and highly reactive liquid that releases toxic gases (HCl and SO₂) on contact with moisture. Exposure may lead to severe respiratory irritation, pulmonary edema, eye damage, and skin burns. Strict moisture-free handling, appropriate PPE, effective ventilation, and immediate decontamination are essential.

a) Threshold Limit Value (TLV-TWA) — No data available for (TLV-TWA) for an 8-hour time-weighted average (TWA)

NOTES

1 Threshold Limit Value Ceiling (TLV-C) — 0.2 ppm (ACGIH).

2 Threshold Limit Value Ceiling (TLV-C) — 1ppm (4.87 mg/m³) (OSHA PEL & NIOSH).

b) Short Term Exposure Limit (STEL) — 1 ppm (4.87 mg/m³) (UK).

c) Immediately Dangerous to life or Health (IDLH) — No data available.

d) Lethal Dose (LD₅₀), (Rat), Dermal — No data available.

e) Lethal Dose LD_{L0} (Rat), Oral — 324 mg/kg.

f) Inhalation (Mouse) Lethal Concentration (LC₅₀-4 hour) — 2.717 mg/l.

5.4 Antidote

Thionyl chloride has no specific antidote. Seek medical help in case of exposure.

5.5 Health Effects

Thionyl chloride vapour irritates respiratory system, eyes and skin. The liquid is highly corrosive and causes burns on the skin, eyes and mucous membranes, whenever it comes in contact with these parts. Its ingestion results in severe internal irritation and damage. However, no permanent ill effect is reported.

5.5.1 Signs and Symptoms

Exposure to thionyl chloride causes conjunctivitis, dermatitis, rhinitis (inflammation of nose) and pneumonia primarily through inhalation of fumes or by direct contact with liquid. The compound may react with moisture to release corrosive gases such as hydrogen chloride (HCl) and sulphur dioxide (SO₂), which irritate and damage the

eyes, nose, and throat, causing coughing and shortness of breath. Eye exposure may lead to intense pain, redness, tearing, and blurred vision, while skin contact can cause irritation, redness, blistering, and severe chemical burns.

5.5.2 Acute Toxicity

5.5.2.1 Systemic effects

Acute exposure to thionyl chloride can cause irritation of the respiratory mucous membranes, burning of the eyes, nose, and throat, coughing, and shortness of breath. High concentrations may lead to respiratory distress, central nervous system depression, dizziness, nausea, loss of consciousness, or coma. Ingestion causes severe corrosive injury to the gastrointestinal tract and may result in chemical pneumonitis. Skin contact causes irritation and damage, while eye contact causes intense pain, redness, tearing, and possible corneal injury.

5.5.2.2 Local effects

Skin contact with thionyl chloride may lead to severe burns, redness, and irritation, and prolonged or repeated exposure increases the risk of secondary infection and systemic absorption through damaged skin. Eye contact may cause intense pain, redness, tearing (lacrimation), and blurred vision, and severe exposure results in corneal damage or conjunctivitis.

5.5.3 Chronic Toxicity

Repeated or prolonged exposure may lead to respiratory complications, chronic irritation of mucous membranes, and potential damage to the liver and kidneys. Neurological effects also include headaches, dizziness, and fatigue.

5.5.3.1 Systemic effects

Chronic exposure to thionyl chloride may result in accumulation in body tissues and may lead to long-term systemic effects.

5.5.3.2 Local effects

Chronic local exposure to thionyl chloride from repeated contact with dilute solutions and may causes superficial skin damage or primary irritant dermatitis. Inhalation of dust or mist causes irritation of the respiratory tissues.

6 PERSONAL PROTECTIVE EQUIPMENT

6.1 Availability and Use

6.1.1 While personal protective equipment is not an adequate substitute for good safe working conditions, adequate ventilation, and intelligent conduct on the part of employees working with thionyl chloride, it complements these safety measures. Also in several instances, it may be the only practical means of protecting the worker, particularly in emergency situations.

6.1.2 The correct usage of personal protective equipment requires the education of the worker in proper employment of the equipment available to personnel.

6.1.3 Under conditions which are sufficiently hazardous to require personal protective equipment, its use should be supervised and the type of protective equipment selected should be capable of control over any potential hazard.

6.2 Non-Respiratory Equipment

6.2.1 Eye and Face Protection

6.2.1.1 Chemical safety goggles

Employees working with thionyl chloride should wear gas-tight chemical goggles cup type or rubber framed goggles equipped with approved impact glass or plastic lenses should be worn, whenever there is danger of thionyl chloride coming in contact with the eyes (*see* IS 8520).

6.2.1.2 Face shields

Plastic shields (*full length, 20cm minimum*) with forehead protection shall be worn over safety protection goggles (see IS 8520).

6.2.1.3 Spectacle-type safety goggles

Metal or plastic rim safety spectacles with unperforated side shields should be used where continuous eye protection is desirable. If person is using corrective lenses he/she shall wear the safety goggles over that. These types, however, should not be used where complete eye protection against thionyl chloride is needed (see IS 8520).

6.2.2 Head Protection

Safety helmets as specified in IS 2925 should be worn.

6.2.3 Foot and Leg Protection

Rubber safety shoes with built-in steel toe caps are recommended for workers handling drums and carboys of thionyl chloride (see IS 10667). Rubbers worn over leather safety shoes [see IS 15298 (Part 2)] should be thoroughly cleaned after contamination.

6.2.4 Body, Skin and Hand Protection

6.2.4.1 Suits made of polyvinyl chloride and neoprene rubber or suitable impervious protective material and properly designed, should be used wherever complete body protection (see IS 8519) is necessary.

6.2.4.2 Aprons made of rubber or other suitable protective material should be used for protection against accidental contact.

6.2.4.3 Affected areas should be thoroughly washed with soap and water, excluding the eyes. To protect against splashes, gloves and aprons made of neoprene or other impervious materials can be worn. These garments should be cleaned inside and outside each time they are contaminated.

6.2.4.4 Sleeves made of suitable protective materials should be worn when the need for complete arm protection is indicated.

6.2.4.5 When cleaning, inspecting, or repairing tanks, safety equipment, such as safety belts, rescue harness, life-line, protective clothing and gas masks should be worn as required by the specific nature of the work and the hazards involved.

6.2.4.6 Frequent inspections and necessary repairs should be made of all personal protective equipment so that the wearer will be adequately protected. Neoprene rubber and other impervious apparel must be checked frequently for signs of deterioration due to exposure to thionyl chloride.

6.2.4.7 All contaminated clothing, including gloves, shoes, coveralls, etc., should be removed as soon as possible after exposure to avoid prolonged contact with thionyl chloride. It should be thoroughly decontaminated and cleaned before re-use.

6.3 Respiratory Equipment

6.3.1 Severe exposure to thionyl chloride may occur when decontaminating areas following spills, or in case of failure of piping or equipment. Employees who may be subject to such exposure should be provided with proper respiratory protection and trained in its use and care. The appropriate respiratory equipments are mentioned in **6.3.2** to **6.3.6**.

NOTE — Respiratory protective equipment shall be carefully maintained, inspected, cleaned and sterilized at regular intervals, and always before and after use by another person.

6.3.2 Self-Contained Breathing Apparatus (SCBA)

Self-contained breathing apparatus (SCBA) permits the wearer to draw air from the cylinder which is the part of the apparatus [see IS 10245 (Part 2)]. SCBA allows considerable mobility. The length of time a self-contained breathing apparatus provides protection varies according to the amount of air, regenerating material carried.

6.3.3 Positive Pressure Hose Masks

Air shall be supplied by blowers requiring no internal lubrication. The wearer shall be able to use the same route for exit as for entrance and shall take precautions to keep the hose line free of entanglement. The air blower shall be placed in an area free of contaminants.

6.3.4 Air Line Masks

This type of devices is supplied with clean compressed air. These masks are supplied with air piped from an air compressor. It is extremely important that the air supply is taken from a safe source and that it is not contaminated by oil decomposition from inadequate cooling at the compressor. The safer method is to use a separate compressor of the type not requiring internal lubrication. Pressure reducing and relief valves as well as suitable traps and filters shall be installed at all mask stations [see IS 10245 (Part 3)].

6.3.5 Full Face Mask

Full face masks used for thionyl chloride handling must comply with IS 14166, ensuring high leak-tightness, durable visor protection, and a secure threaded connector for use with supplied-air or SCBA systems. These masks must provide complete sealing of the eyes, nose, and mouth to prevent exposure to the corrosive fumes (HCl and SO₂) released by thionyl chloride. Class 2 or Class 3 full face masks are recommended, as they offer enhanced material strength, resistance to deformation, and compatibility with high-hazard respiratory equipment.

6.3.6 Canister Type Gas Masks

These devices are equipped with full face piece, fitted with the proper canister (organic vapour), approved for absorbing thionyl chloride vapour. These will offer protection against concentration not exceeding 2 percent by volume when used in accordance with manufacturer's instructions. The mask should be used for relatively short exposure periods only. They may not be suitable for use in an emergency since, at the time, the actual vapour concentration is unknown and an oxygen deficiency may exist. The wearer shall be warned to leave the contaminated area immediately on detecting the odour of a harmful vapour. This may indicate that the mask is not functioning properly, that the vapour concentration is too high, that the canister is exhausted or that the mask is not properly fitted.

CAUTION:

- 1 Filter type respirators do not offer protection against gases and are unsuitable for use when working with thionyl chloride.
- 2 The oxygen content of the air in the work environment shall not be less than 17 percent by volume.

NOTE — Respiratory protective equipment shall be carefully maintained, inspected, cleaned, and sterilized at regular intervals and always before and after use by another person.

7 STORAGE, HANDLING, LABELLING AND TRANSPORT

7.1 General

7.1.1 All personnel handling thionyl chloride should use proper personal protective equipment. Appropriate firefighting equipment should be available in the vicinity while handling thionyl chloride. Persons handling thionyl chloride should have adequate training in use of firefighting equipment.

NOTE — The storage shall be as per the criteria given in the relevant schedule of the *Manufacture, Storage and Import of Hazardous Chemicals Rules*, 1989. While referring to the statutes, the stipulations given in the latest amendments of those statutes shall be taken into account.

7.1.2 The manufacturer/producer of thionyl chloride shall issue material safety datasheet (MSDS) for all its users for comprehensive safety details. The MSDS shall be in the format as prescribed in IS 17889. MSDS translation in local languages may be provided as required.

7.2 Storage

7.2.1 Depending on the quantity, thionyl chloride shall be stored in glass bottles with inner cushioning and absorbent material when packed together in wooden boxes.

7.2.2 In the case of bulk quantities, it shall be stored in glass carboys covered with closely fitting wicker or bamboo work, packed individually in a wooden slatted crate iron hamper.

7.2.3 It may also be stored in suitable heavy duty galvanized drums with tightly closed bungs.

7.2.4 The packed material shall be stored in cool, dry and sufficiently ventilated rooms. A standard acidic liquids storage room or cabinet is sufficient.

7.2.5 The material shall be stored away from water.

7.2.6 The containers shall be handled carefully while loading, transporting or unloading, as damaged containers may leak and cause dangerous concentrations of its vapour in an enclosed space.

7.2.7 Unreacted thionyl chloride escaping from the reaction vessel shall be scrubbed with caustic soda solution or soda ash.

7.2.8 In case of fire-fighting, water shall not be used. Only dry powder or carbon dioxide fire extinguisher should be used.

7.2.9 Isolate the material from incompatible materials such as water, ammonia and other bases, alcohols, amines, most common metals, dimethyl sulfoxide, chloryl perchlorate, linseed oil, quinoline, sodium, 2,4-hexadiyn-1-6-diol, o-nitrobenzoylactic acid, and onitrophenylacetic acid.

7.2.10 Due to the hazardous nature of the material only minimal quantities of material should be purchased and stored.

7.3 Handling

7.3.1 General

Each consignment of thionyl chloride should be unloaded carefully to avoid impact, tipping or damage to containers. Inspect every package and container immediately on receipt for dents, distortion, corrosion or signs of leakage. If leakage is found, it should be handled with particular care to prevent further leakage and removed to a safe place where the leak can be stopped and contents transferred to another container. The general handling practices given in 7.3.1.1 to 7.3.1.9 should be followed.

7.3.1.1 All personnel shall be trained on the hazards, proper handling, storage, use of PPE, and emergency procedures related to thionyl chloride before commencing any work.

7.3.1.2 Areas where thionyl chloride is handled shall be well ventilated. high standard of plant maintenance shall be maintained to ensure hygienic working conditions.

7.3.1.3 Adequate protective safety equipment such as PVC gloves, PVC aprons, goggles, face shield canister gas mask (preferably, compressed air mask), etc, should be worn when handling thionyl chloride.

7.3.1.4 Before opening a container, complete protective clothing, goggles and effective respirator or positive air supply hood shall be worn. Clean, preferably white, PVC gloves shall be used as rubber absorbs thionyl chloride after prolonged use.

7.3.1.5 Prevent all contact with water, moisture, alcohols, strong acids, strong bases, amines, ammonia, reactive metals, DMSO, and DMF, as these can cause violent or exothermic reactions.

7.3.1.6 Thionyl chloride shall be stored in tightly sealed, compatible containers in a cool, dry, and well-ventilated area. Keep away from heat sources, as thionyl chloride decomposes above 140 °C, releasing corrosive gases (HCl and SO₂).

7.3.1.7 Explosion-proof electrical equipment and fittings shall be used.

7.3.1.8 Conduct all manipulations of thionyl chloride inside a properly functioning chemical fume hood, glove box, or certified local exhaust system vented directly outdoors.

7.3.1.9 Handling in such controlled systems is less hazardous than storage in glass carboys.

7.4 Labelling

7.4.1 Each container (including tankers) shall carry an identifying label or stencil as specified in Fig. 15 of IS 1260 (Part 1) or Label Model No. 8 of Annex B of IS 18149. The storage containers shall be labelled or marked to identify as follows:

- a) Name of the material;

- b) Indication of the source of manufacture;
- c) Net mass;
- d) Batch number or lot number in code or otherwise; and
- e) Physical, chemical and toxicological data as per the criteria given in the relevant schedule of the *Manufacture, Storage and Import of Hazardous Chemicals Rules*, 1989. While referring to the statutes, the stipulations given in the subsequent amendments of those statutes shall be taken into account.

7.4.2 All containers of thionyl chloride shall bear an identifying label in depicted Fig. 15 of IS 1260 (Part 1). (Thionyl chloride has been classified as corrosive as per IS 1446. The containers shall in addition be labelled as follows:

THIONYL CHLORIDE

WARNING: CAUSES BURNS, VAPOUR IRRITATING
KEEP THE CONTAINER CLOSED IN A COOL PLACE
AVOID CONTACT WITH EYES, SKIN AND CLOTHING
DO NOT ALLOW WATER TO GO INTO CONTAINER
REACTS VIOLENTLY WITH WATER
USE WITH ADEQUATE VENTILATION

7.4.3 Containers of waste or other materials contaminated with thionyl chloride should be labelled as follows:

THIONYL CHLORIDE CONTAMINATED MATERIAL
DISPOSE OFF OR DECONTAMINATE USING
APPROVED PROCEDURE

7.5 Transport

7.5.1 General

7.5.1.1 These vehicles must be properly ventilated to prevent the accumulation of toxic vapours and, wherever practicable, should be leak-proof.

7.5.1.2 Vehicles transporting thionyl chloride must avoid overcrowded residential areas when possible, and transport should comply with local and national safety standards for hazardous materials.

7.5.2 Precautions During Transport

7.5.2.1 Containers shall be tightly closed, properly labelled/marked, and secured upright. Use secondary containment during transport/storage to contain leaks and prevent mixing with incompatible materials (water, alcohols, amines, bases, metals).

7.5.2.2 Drivers and personnel handling thionyl chloride must be trained in hazard recognition, safe handling, and emergency procedures.

7.5.2.3 Transport routes should be planned to minimize environmental exposure in the event of accidental spillage.

7.5.3 Driver

7.5.3.1 Vehicle driver should be aware of the potential hazards of the load and should know standard procedure (do's and don'ts) in the event of an accident or an emergency.

7.5.3.2 The driver should be familiar with the truck and the ISO-container; it's length, width, maximum speed, height, braking, turning circle etc.

7.5.3.3 The driver should carry TREM card when vehicle is on road.

7.5.3.4 The driver should not board the passengers in transport units carrying dangerous goods.

7.5.3.5 The Driver and his crew should be aware about the necessary requirements for fire-fighting.

7.5.3.6 The drivers should be prohibited from opening the packages containing dangerous goods.

7.5.3.7 The engine shall be shut off during loading and unloading operations.

NOTE — If transport of the hazardous chemical is involved it shall be carried out in accordance with the *Central Motor Vehicles Rules*, 1989. While referring to the statutes, the stipulations given in the subsequent amendments of those statutes shall be taken into account.

8 SPILLAGE, LEAKAGE AND WASTE DISPOSAL

8.1 General

8.1.1 Potential areas of leakage shall be inspected regularly. Environmental monitoring should be carried out at different workplaces regularly to detect thionyl chloride concentration. Complete and accurate records of all environmental measurements should be maintained.

8.1.2 *Determination in the Atmosphere*

The concentrations of thionyl chloride vapour in the air shall be determined by drawing contaminated air through two scrubbers containing a measured quantity of 0.1 N silver nitrate solution acidified with nitric acid. After sampling, 0.1 N sodium chloride solution equivalent to the silver nitrate used shall be added, and the excess chloride shall be back titrated with 0.1 N silver nitrate.

8.2 Spillage

8.2.1 General

8.2.1.1 In case of spills, the affected area shall be covered with sodium bicarbonate or an equal mixture of soda ash and slaked lime. After thorough mixing, water shall be sprayed from an atomizer with adequate precautions. This mixture should then be transferred slowly to a large container of water. It is then neutralized and drained into the sewer with sufficient water.

8.2.1.2 Solid waste material containing significant level of residual thionyl chloride should be transferred from the plant to special containers located in designated areas within the works boundary where it shall be suitably treated.

8.2.2 *Land Spill (Spill on land)*

8.2.2.1 *Containment*

8.2.2.1 In the case of spill, evacuate the danger area and restrict access to authorized personnel only. Consult a hazardous materials expert where necessary.

8.2.2.2 Entry of thionyl chloride into sewers, basements, confined areas, or natural water sources shall be prevented. The spread shall be controlled by diking or constructing impervious barriers using non-reactive materials. Dry, inert absorbents such as vermiculite, dry sand, or earth shall be used to confine and collect the liquid, and unnecessary disturbance of the material shall be avoided to minimize vapour generation.

8.2.2.3 All ignition sources, including open flames, heat sources, and smoking materials, shall be eliminated.

8.2.2.4 Avoid exposure of water, alcohols, bases, amines, oxidizers, and reactive metals to thionyl chloride due to its highly reactivity.

8.2.2.5 Isolate the spill or leak area for at least 50 m in all directions. If a tank, rail car, or highway tank is involved in fire, isolate the area for 800 m in all directions and consider initial evacuation for the same distance.

8.2.2.2 *Consequence*

In case of exposure, thionyl chloride shall rapidly hydrolyse on contact with moisture in air, soil, or water to form hydrogen chloride (HCl) and sulphur dioxide (SO₂). The corrosive and toxic gases formed may produce hazardous vapour clouds causing severe respiratory irritation, chemical burns to the skin and eyes, acidification of nearby soil and surface water, and toxicity to aquatic life. Residual liquid may corrode container materials and may be absorbed into porous soil, where continued hydrolysis shall generate acidic runoff and vapours.

8.2.2.3 *Mitigation*

Only trained personnel equipped with appropriate respiratory protection (positive-pressure SCBA or supplied-air respirator) and acid-resistant PPE shall enter the affected area. Ignition sources shall be removed and contact with incompatible materials such as water, alcohols, bases, amines, and metals shall be prevented. Contaminated absorbents and residues shall be collected in compatible, tightly sealed and labelled containers for hazardous waste disposal. Mechanical aids shall be used to minimize manual handling, and the area shall be ventilated from upwind locations using suitable exhaust or scrubber systems to remove HCl and SO₂ vapour. Neutralization or gas scrubbing shall be carried out only in controlled systems by trained personnel using appropriate engineering controls.

8.2.3 Water Spill (*Spill in Water*)

Avoid release of thionyl chloride into water, as it reacts vigorously with water to produce corrosive hydrogen chloride (HCl) and sulphur dioxide (SO₂) gases with heat evolution. Entry into surface waters or sewers shall be prevented, as it may rapidly generate toxic and acidic conditions hazardous to aquatic life, infrastructure, and downstream users.

8.2.3.1 Containment

In case of a water spill, isolate the affected area immediately and stop the source of discharge. The concerned environmental authorities shall be notified immediately, and downstream water quality shall be continuously monitored to ensure effective containment and remediation. Dilution with large volumes of water shall be avoided, as it accelerates hydrolysis and gas generation, and measures that may disperse the material or increase vapour formation shall be avoided.

8.2.3.2 Mitigation

Only trained personnel shall be allowed to enter the affected area. Contaminated water, residues, and wash waters shall be collected and treated in controlled facilities or sent to authorized hazardous-waste processors. Discharge into sanitary or storm sewers shall not be permitted without prior approval from the competent authorities.

8.3 Leakages

8.3.1 General Information

In case of leakage, the affected area shall be covered with sodium bicarbonate or an equal mixture of soda ash and slaked lime. After thorough mixing, water shall be sprayed using an atomizer with adequate precautions. The mixture shall then be transferred slowly to a large container of water, neutralized, and discharged into the sewer with sufficient dilution.

8.3.2 Leak from the Truck

For minor leaks, a compatible chemical-resistant sealant shall be applied where it can be done safely from an upwind position. In case of a major leak, the material shall be transferred to another compatible tanker or to sealed barrels or drums using closed and grounded transfer systems. Dripping leaks shall be collected in smaller compatible containers placed beneath the leak point, ensuring no contact with water or moisture.

CAUTION

All ignition sources, such as smoking materials, sparks, flares, or open flames, shall be eliminated from the immediate area. Inhalation of corrosive vapour such as hydrogen chloride (HCl) and sulphur dioxide (SO₂) shall be avoided. Grounded and bonded metal pipelines shall be used during transfer operations to prevent static electricity accumulation. Contact of thionyl chloride with water, moisture, alcohols, amines, bases, or reactive metals shall be strictly avoided due to the risk of violent reactions.

8.4 Waste Disposal

8.4.1 General

In case of spills and leakage the affected area shall be covered with sodium bicarbonate or an equal mixture of soda ash and slaked lime. After thorough mixing, water shall be sprayed from an atomizer with adequate precautions. This mixture should then be transferred slowly to a large container of water. It is then neutralized and drained into the sewer with sufficient water.

8.4.1.1 All local and state regulations concerning waste disposal should be complied with.

8.4.1.2 Waste thionyl chloride shall be disposed of in compliance with as per *Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016*.

8.4.1.3 Solid waste material containing significant level of residual thionyl chloride shall be transferred from the plant to special containers located in designated areas within the works boundary, where it shall be suitably treated.

8.4.1.4 All residual materials and rinsate from empty containers of this material shall be collected and disposed of as hazardous waste.

8.4.1.5 The restate from decontamination of all non-disposable equipment shall be collected and disposed as hazardous waste.

8.4.1.6 All disposable materials contaminated with this material must be disposed as hazardous waste.

8.4.1.7 A chemical pick-up request form must be completed and submitted when the hazardous waste needs to be removed.

8.4.1.8 Residues containing thionyl chloride should be disposed as per *Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016*.

9 FIRE PREVENTION AND FIRE FIGHTING

9.1 General

9.1.1 Thionyl chloride is non-flammable; however, exposure to heat or fire causes decomposition and releases highly toxic and corrosive gases such as hydrogen chloride (HCl), sulphur dioxide (SO₂), chlorine, and phosgene.

9.1.2 Fire involving nearby combustible materials heats thionyl chloride containers and may cause pressure build-up, container rupture, and sudden release of corrosive vapours.

9.1.3 The toxic and acidic fumes produced during fire decomposition pose serious hazards to human health and the environment and may cause severe respiratory irritation, chemical burns, and atmospheric corrosion.

9.2 Prevention

9.2.1 Thionyl chloride shall be stored away from water, moisture, strong bases, alcohols, amines, reactive metals, and oxidizing substances to prevent violent or exothermic reactions.

9.2.2 Areas used for storage of thionyl chloride shall be cool, dry, and well ventilated. A high standard of plant maintenance shall be maintained to ensure safe and hygienic working conditions.

9.2.3 Use spark-proof, grounded, and bonded equipment during transfer operation to prevent static discharge. Containers made of compatible and corrosion-resistant materials shall be used.

9.2.4 Smoking, open flames, welding operations, and other ignition sources shall be strictly prohibited in areas where thionyl chloride is stored or handled.

9.2.5 Suitable fire detection systems shall be installed, and appropriate fire extinguishers such as dry chemical or carbon dioxide units shall be readily available for controlling fires involving surrounding materials.

9.2.6 Containers shall be clearly labelled, access shall be restricted to trained personnel, PPE requirements shall be enforced, and periodic safety inspections shall be conducted to ensure compliance with storage and fire-prevention measures.

9.2.7 Adequate personal protective equipment such as PVC full body suit, PVC gloves/Viton(R)/ FKM (fluoro rubber)/ Butyl rubber gloves, goggles, face shield canister gas mask (preferably, compressed air mask), etc, shall be worn while handling thionyl chloride.

9.2.8 Before opening a container, full protective clothing, goggles and an effective respirator or positive air supply hood shall be worn. Clean, preferably white, PVC gloves shall be used as rubber absorbs thionyl chloride after prolonged use.

9.3 Fire Fighting

9.3.1 Avoid direct application of high-pressure water on thionyl chloride, as contact with water causes violent hydrolysis and release of corrosive fumes. Dry chemical, carbon dioxide, or alcohol-resistant foam shall be used to extinguish fires involving surrounding materials. Water spray may be used only from a safe distance to cool unopened containers.

9.3.2 Firefighters shall wear positive-pressure self-contained breathing apparatus (SCBA) and full chemical-protective clothing, including acid-resistant suits, gloves, and face protection, due to the presence of toxic decomposition vapours.

9.3.3 Contaminated runoff shall be prevented from entering sewers, drains, or natural water bodies. The runoff shall be collected, contained, and disposed of as hazardous waste.

9.3.4 The area shall be monitored for residual hydrogen chloride (HCl) and sulphur dioxide (SO₂) vapours during and after firefighting operations. Re-entry shall be permitted only after air monitoring confirms safe concentrations.

10 TRAINING

10.1 Safety in handling thionyl chloride depends upon the effectiveness of employee education, training and supervision. The education and training of employees to work safely and to use the personal protective equipment and other safe guards provided for them is a responsibility of supervision. Employee education and training should emphasize the need of handling thionyl chloride according to the methods outlined in the manual, in order to avoid spilling or splashing, leaks, burns, inhalation of the vapour of burning material, or ingestion. Unauthorised and untrained employees should not be permitted in areas where thionyl chloride is being handled. The work permit shall be issued as per format prescribed in IS 17893.

10.2 Before being placed on the job, all new employees should be instructed thoroughly for the proper handling of thionyl chloride. Older employees should be re-instructed periodically. Instructions and periodic drill or quiz regarding the locations, purpose and use of personal protective equipment.

10.3 Each employee should know the location, purpose and maintenance of personal protective equipment (PPEs) and be thoroughly trained when and how to use it. Each employee should know the location of safety showers, fountains for flushing the eyes, and hose lines.

10.4 Only reliable, dependable and properly trained employees should be given the responsibility of all operations involving storage, handling, transport and emergency management involving thionyl chloride.

10.5 Employee should be trained to report to the proper authority about all suspected leaks or equipment failure.

10.6 Employee shall be instructed to inform supervisor regarding any signs of illness or skin related problems.

10.7 Each employee should know what to do in emergency and the first-aid measures and should realize the necessity for the prompt application of first-aid in case of contact with thionyl chloride or exposure to the vapours of the burning material.

10.8 Employee education and training should include instruction and periodic drill or quiz regarding the locations of safety showers, eye-baths, bubbler drinking fountains, or the closest source of water for use in emergencies.

10.9 Employee education and training should include instructions to avoid all unnecessary inhalation of vapours of thionyl chloride and all direct contact with the liquid.

10.10 Employee education and training should include instructions to report to the proper authority all signs of illness. Preplacement training should include information about safety procedure and action in emergency; and refresh training should give periodically.

10.11 Employee education and training should include instructions to report to the proper authority all cases of equipment failure.

10.12 Employee education and training should include instruction concerning the synergistic effect of alcohol ingestion in this type of toxicity.

10.13 Workers should be thoroughly informed of the hazards that may result from improper handling of thionyl chloride. They should be cautioned to prevent spills and thoroughly understand that any spills shall be cleaned up immediately. Each employee should know what to do in an emergency and should be fully informed as to first-aid measures.

11 HEALTH MANAGEMENT, FIRST-AID AND MEDICAL TREATMENT

11.1 Health Monitoring

Potential areas of leakage shall be inspected regularly. Environmental monitoring shall be carried out at different work places regularly to detect thionyl chloride concentration. Complete and accurate records of all environmental measurements shall be maintained.

11.1.1 Personal Hygiene

11.1.1.1 Hands, face, and exposed skin shall be washed thoroughly with soap and water before meals or breaks. After completion of work, workers shall wash the face, hands, forearms, and hair, and a full-body wash shall be taken before leaving the facility.

11.1.1.2 Employees handling thionyl chloride shall bathe immediately after work. Symptoms such as coughing, breathing difficulty, throat irritation, eye irritation, or accidental skin contact shall be reported immediately to the medical department. Consumption of alcohol after exposure shall be avoided.

11.1.1.3 Food shall not be stored, handled, or consumed in areas where thionyl chloride is used. Eating, drinking, smoking, or applying cosmetics shall be permitted only in designated clean areas after proper hand washing.

11.1.1.4 Contaminated work clothing shall be removed immediately and laundered on the premises before reuse. Separate lockers shall be provided for work clothing and personal clothing to prevent cross-contamination.

11.1.1.5 Emergency showers and eye-wash facilities shall be installed in all areas where exposure to thionyl chloride may occur (see IS 10592). Workers shall be trained in the location and proper use of this equipment.

11.1.2 Determination in the Atmosphere

Concentrations of thionyl chloride vapour in the air shall be determined by drawing contaminated air through two scrubbers containing a measured quantity of 0.1 N silver nitrate solution acidified with nitric acid. When sampling has been completed, 0.1 N sodium chloride solution equivalent to the silver nitrate used shall be added and the excess chloride back titrated with 0.1 N silver nitrate.

11.1.3 Physical Examination

11.1.3.1 Pre-Placement physical examinations

Pre-placement medical examination shall be conducted to determine the fitness of workers for handling corrosive and toxic chemicals. Individuals suffering from asthma, chronic respiratory disease, severe allergies, or serious skin disorders shall not be assigned to work involving thionyl chloride. Baseline respiratory function tests, eye examination, and a general physical assessment shall be conducted.

11.1.3.2 Periodic examination

11.1.3.2.1 Periodic medical examinations shall be conducted at regular intervals, normally every 6 to 12 months, in accordance with occupational health requirements. These examinations shall include respiratory function tests, skin assessment, and evaluation of exposure-related symptoms.

11.1.3.2.2 Workers reporting respiratory irritation, coughing, eye discomfort, or repeated skin irritation shall undergo immediate medical evaluation. Continuous health monitoring shall be carried out to detect early signs of chronic respiratory sensitivity or other adverse health effects associated with exposure.

11.2 First Aid

11.2.1 General Principles

Remove soaked clothing and wash affected skin with soap and water. Wash contaminated clothing with water before reusing. In case of serious burns call for medical attention immediately. Carry affected person on a stretcher to the medical department or first aid station.

11.2.2 *Contact with Skin*

Remove all contaminated clothing. The affected person shall be placed under an emergency shower and the affected skin shall be washed thoroughly with soap and lukewarm water.

11.2.3 *Contact with Eyes*

If thionyl chloride has got into the eyes wash them with plenty of water. A competent physician should be consulted as early as possible.

11.2.4 *Ingestion*

In case of ingestion, wash out the mouth thoroughly with water and give plenty of water to drink, followed by milk of magnesia or 5 percent solution of sodium bicarbonate may be used as well as bronchodilators and decongestants. Obtain medical attention.

11.2.5 *Inhalation*

In case of inhalation of thionyl chloride, the affected person should be removed to a clean environment and allow to rest, keeping him warm. In severe cases oxygen should be administered using intermittent positive pressure breathing apparatus, and also immediate medical attention should be obtained.

11.2.6 *First-Aid Kit*

In the case of thionyl chloride exposure, first aid involves immediate flushing of the skin with plenty of water for at least 15 min to 20 min; however, rinsing the mouth is not advised due to its violent reaction with water.

12 ADDITIONAL INFORMATION

12.1 Tank and Equipment Cleaning and Repairs

12.1.1 *Preparation of Tank and Equipment*

12.1.1.1 Inspection, cleaning, and repair of tanks and associated equipment shall be carried out only under an authorized work permit and under the direct supervision of trained personnel familiar with the hazards of thionyl chloride and the necessary safety precautions. The principal hazards include exposure to corrosive liquid, toxic vapours, and violent reactions with moisture.

12.1.1.2 Tanks, pumps, pipelines, and valves shall be completely drained before any repair work is undertaken. Repair work shall not commence until the equipment is confirmed to be free from thionyl chloride liquid and vapours. When opening flanges, the lower and farthest bolts shall be loosened first to control residual drips and to minimize direct exposure.

12.1.1.3 Pipelines connected to the tank shall be physically disconnected by removing a section of the line and installing blank flanges on the open ends to prevent accidental leakage or ingress. Isolation by valves alone shall not be relied upon.

12.1.1.4 Residual liquid shall be removed using dry and compatible absorbent materials. Washing the tank with water shall not be permitted, as thionyl chloride reacts violently with water releasing hydrogen chloride and sulphur dioxide. Only approved neutralizing or cleaning agents suitable for moisture-reactive chlorinating agents shall be used.

12.1.1.5 If steaming is required as part of the decontamination process, it shall be carried out only after confirming that the equipment is free from bulk thionyl chloride. The operation shall be performed under controlled conditions using a closed scrubbing system to capture and neutralize the acidic gases produced. Condensate shall be continuously drained to a neutralization system.

12.1.1.6 After completion of cleaning, the tank shall be ventilated and allowed to cool. The absence of thionyl chloride vapours shall be verified using suitable air-monitoring equipment before permitting personnel entry.

12.1.2 *Entering Tank/Vessel*

12.1.2.1 A tank, vessel, or any confined space shall not be entered without a valid Confined Space Entry Permit confirming that the equipment has been properly cleaned, ventilated, tested, and declared safe for entry.

12.1.2.2 The entry opening shall remain unobstructed to permit emergency exit. Personnel entering the tank shall wear suitable respiratory protection such as self-contained breathing apparatus (SCBA) or supplied-air respirator and shall be secured with a safety lifeline monitored by personnel stationed outside.

12.1.2.3 At least one trained attendant shall remain outside the tank and maintain continuous communication with the person inside. At least two additional trained personnel shall be available nearby to provide immediate assistance in case of emergency.

12.1.2.4 Continuous forced ventilation and supply of fresh air shall be provided during cleaning, inspection, or repair operations to prevent accumulation of acidic vapours such as hydrogen chloride (HCl) and sulphur dioxide (SO₂).

12.1.2.5 For tanks having openings only at the top, complete removal of vapours shall be ensured before entry. Recirculation of exhaust air back into the tank during ventilation shall be prevented.

12.1.3 *Repairs*

12.1.3.1 Repair operations such as cutting, welding, hot work, riveting, or grinding shall not be carried out unless a valid Hot-Work Permit and Confined Space Entry Permit have been issued by an authorized person confirming the absence of thionyl chloride vapours.

12.1.3.2 If repair work is interrupted for more than one hour, the permits shall be revalidated and the tank atmosphere shall be retested before resumption of work.

ANNEX A

(Clause 2)

<i>IS No.</i>	<i>Title</i>
IS 1446 : 2024	Dangerous Goods — Classification (<i>third revision</i>)
IS 11305 : 1985	Specification for thionyl chloride
IS 1260 (Part 1) : 1973	Pictorial markings for handling and labelling of goods Part 1 Dangerous goods (<i>first revision</i>)
IS 2925 : 1984	Specification for industrial safety helmets (<i>second revision</i>)
IS 4155 : 2023	Glossary of terms relating to chemical and radiation hazards and hazardous chemicals (<i>first revision</i>)
IS 4167 : 2020	Glossary of terms relating to air pollution (<i>second revision</i>)
IS 8519 : 2024	Guide for selection of occupational protective clothing — Body protection (selection, care, and maintenance) (<i>first revision</i>)
IS 8520 : 2023/ ISO 19734 : 2021	Eye and face protection — Guidance on selection, use, and maintenance (<i>first revision</i>)
IS 10245 (Part 2) : 2023	Respiratory protective devices — Specification Part 2 Self-contained open circuit breathing apparatus (<i>second revision</i>)
IS 10245 (Part 3) : 2025	Respiratory Protective devices Part 3 Fresh air hose and compressed air line breathing apparatus (<i>second revision</i>)
IS 10592 : 2018	Industrial emergency showers, eye and face fountains and combination units — Specification (<i>first revision</i>)
IS 10667 : 2025	Selection of industrial safety equipment for protection of foot and leg — Guide (<i>first revision</i>)
IS 14166 : 2024	Respiratory protective devices — Full face masks — Specification (<i>first revision</i>)
IS 15298 (Part 2) : 2024	Personal protective equipment Part 2 Safety footwear (<i>third revision</i>) (MOD ISO 20345 : 2021)
IS 17889 : 2022	Material safety data sheets — Guidelines
IS 17893 : 2023	Work permit system — Code of practice
IS 18149: 2023	Transportation of dangerous goods — Guidelines